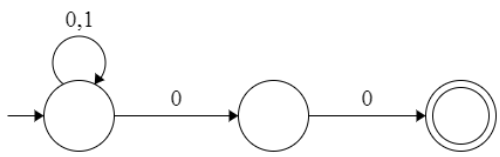
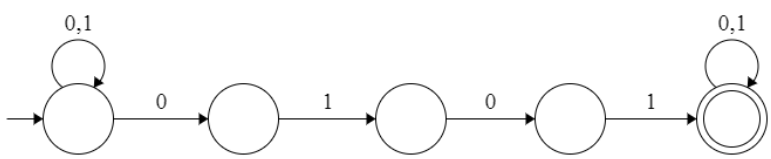
Non-deterministic Finite-state Automata

1. Give state diagrams of NFAs with the specified number of states recognising each of the following languages. In all cases, the alphabet is {0,1}.

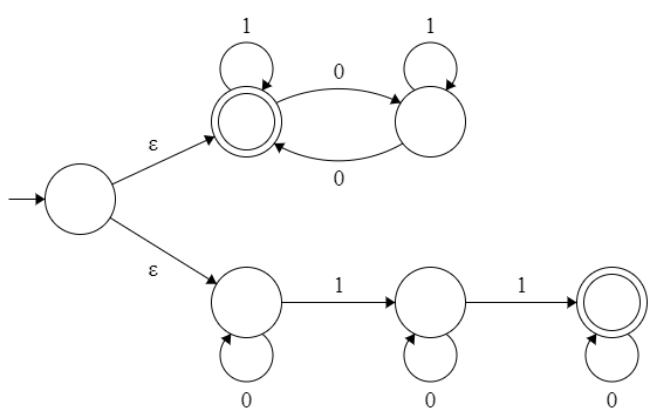
   (a) The language $\{w|w$ ends with 00$\}$ with three states



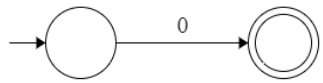
   (b) The language $\{w|w$ contains the substring 0101$\}$ with five states



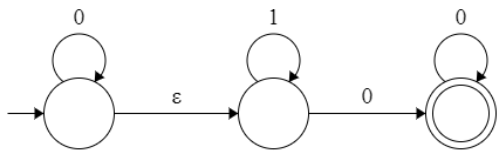
   (c) The language $\{w|w$ contains an even number of 0s, or contains exactly two 1s$\}$ with six states



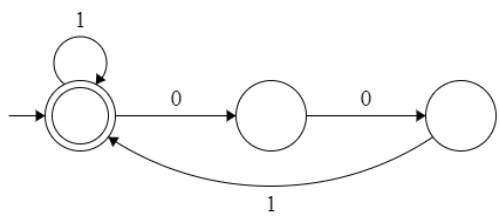
(d) The language $\{0\}$ with two states



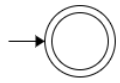
(e) The language $0^*1^*0^+$ with three states



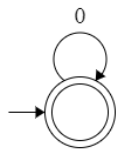
(f) The language $1^*(001^+)^*$ with three states



(g) The language $\{\varepsilon\}$ with one state



(h) The language 0^* with one state

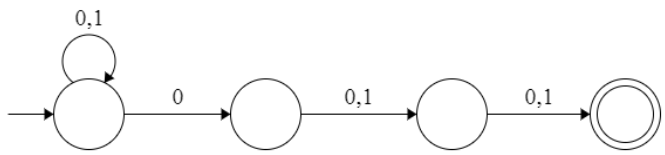


2. Let L_3 be the language over $\{0, 1\}$

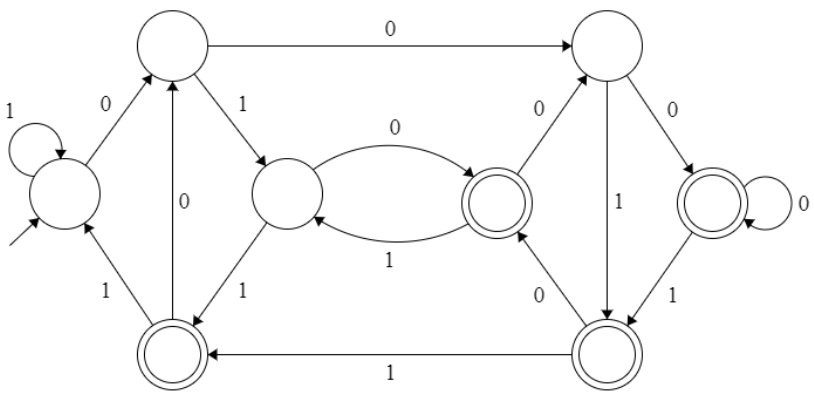
$$L_3 = \{w | \text{ the third letter from the end of } w \text{ is a } 0\}$$

(a) Design an NFA recognising L_3.

Exercise 3 is a special case of Exercise 4. See the solution to Exercise 4 for the logic behind the design of the DFA.



(b) Design a DFA recognising L_3.



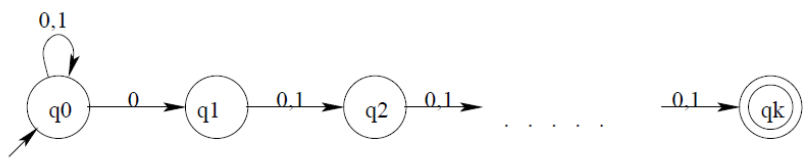
The Answers of Exercises 3 and 4 are from the CS172 Computability and Complexity course at Berkeley.

3. Let L_k be language over $\{0, 1\}$

$$L_k = \{w | \text{ the } k\text{th letter from the end of } w \text{ is a } 0\}$$

(a) Design an NFA recognising L_k.

An NFA for L_k is as shown. At each step, if it sees a 0 it "guesses" if this is in fact the kth last symbol by going into state q_1; from there, it verifies its guess by moving in exactly $k - 1$ more steps to the final state. This NFA has $k + 1$ states.



(b) Design a DFA recognising L_k.

We construct a DFA such that each of the states represents one of the 2^k possible length-k suffixes. Exactly half of these states, those representing suffixes that start with 0, are accepting states. (If a string ends up in one of the other states, the string must have had a 1 in the kth position from the end, and therefore must be rejected.) We transition from state to state by simply lopping off the first digit of the suffix and appending the next character to be read. (For instance, if $k = 4$ then one transition is $(s_{1001}, 0) \to s_{0010}$.)

$$\begin{aligned}
Q &= \{s_z : z \in \{0,1\}^k\} \\
\Sigma &= \{0, 1\} \\
\delta &= \{(s_{aw}, b) \to s_{wb} : a, b \in \{0,1\}, w \in \{0,1\}^{k-1}\} \\
q_0 &= s_{1^k} \\
F &= \{s_z : z \text{ starts with } 0\}
\end{aligned}$$

Note the choice of initial state $q_0 = s_{1^k}$. This is in line with the fact that no zero has been seen when the computation begins.

4. (Optional) Prove that any DFA recognising L_k must have at least 2^k states.

(c) Let M_k be any DFA that accepts L_k. Let us call two strings x and y *distinguishable by* L_k if there is a string w such that *exactly one* of xw and yw is in L_k (i.e., either $xw \in L_k$ and $yw \notin L_k$ or vice versa).

The point about distinguishable strings is the following. We claim that for any pair of distinguishable strings x and y, M_k must end up in *different* states after processing x and y. We can prove this by contradiction. Supose on the contrary that x and y are distinguishable but $\delta^*(q_0, x) = \delta^*(q_0, y)$.[1] Then clearly for any symbol a we must have $\delta^*(q_0, xa) = \delta^*(q_0, ya)$, and repeating, for any string w we have $\delta^*(q_0, xw) = \delta^*(q_0, yw)$. (We could prove this formally by a simple induction on the length of w, with $|w| = 1$ as the base case.) Therefore, it must be the case that M_k either accepts both xw and yw, or rejects them both. So x and y are not distinguishable, a contradiction.

[1] Here δ^* is the extended transition function defined in the obvious way: for a state q and string w, $\delta^*(q, w)$ is the state of the DFA after reading string w starting in state q. Formally, we define $\delta^*(q, a) = \delta(q, a)$ for each symbol a, and $\delta^*(q, wa) = \delta(\delta^*(q, w), a)$.

Now we show that the language L_k has many distinguishable pairs of strings. Consider any two length-k strings x, y with $x \neq y$. Since $x \neq y$, they differ in at least one position: so suppose without loss of generality that $x_i = 0$, $y_i = 1$. Let $w = 1^{i-1}$. Then $xw \in L_k$ and $yw \notin L_k$. (This is because in xw the kth letter from the end is x_i, and similarly for yw.) Thus x and y are distinguishable.

Putting the previous two paragraphs together, we conclude that for any two length-k strings, M_k must end up in different states. Since there are 2^k such strings, M_k must have at least 2^k states.